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II Semester M.C.A. Examination, November/December 2022
(CBCS) (2020 – 21 and Onwards)
COMPUTER SCIENCE
2MCA5 : Design and Analysis of Algorithms

Time : 3 Hours Max. Marks : 70

Instructions : *1) Section – **A** : Answer **any five** questions.*
*2) Section – **B** : Answer **any four** questions.*

SECTION – A

Answer **any five** questions. **Each** question carries **six** marks. **(5×6=30)**

1. Give an account of basic efficiency classes.
2. Present Brute-force based algorithm to perform pattern matching and investigate its efficiency.
3. Develop an algorithm to find whether a list containing integers is sorted or not. Comment on its efficiency.
4. Write an algorithm to evaluate the polynomial.
   $p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_2 x^2 + a_1 x + a_0$ at a given value of x, using straight forward approach.
5. Write dynamic programming based algorithm to compute binomial coefficients. Trace it to compute ${}^7C_3$.
6. Given a digraph, how do you arrive at its transitive closure for the graph ? Write the procedure for the task.
7. Give an account of transform and conquer technique and its variations.
8. Present an algorithm to search for a string in a list of strings using brute force based approach.

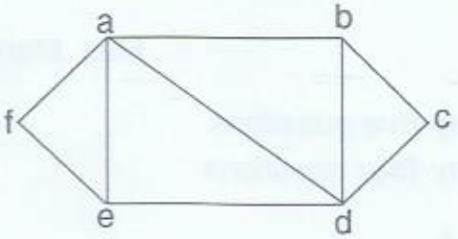
SECTION – B

9. a) Give an account of theoretical analysis of algorithms. **(4×10=40)**
   b) Sort the following list of numbers
   13, 25, 9, 72, 5, 24, 7 using selection sort.

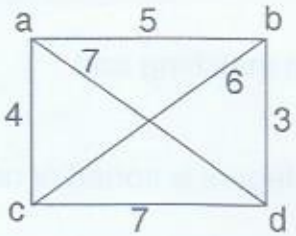
P.T.O.

10. a) Give an account of decrease and conquer technique with an illustrative example for each variation of the technique.

    b) Produce depth-first-search algorithm and comment on its efficiency. Trace it for the following graph.



11. a) Give an account of the underlying principle behind Prim's algorithm. Also produce Prim's algorithm. Trace it for the following graph.



    b) Write transform and conquer based algorithm to evaluate a polynomial. How does it differ from Brute-force based algorithm ?

12. a) Trading space for time is viable in some situations. Justify.

    b) Write Horspool's algorithm to perform pattern matching. How does it differ from Brute-force based pattern matching algorithm ? Comment.

13. a) Write the principle of Backtracking technique. How does it help in solving 4-queens problem ? Explain.

    b) How do you solve sum of subsets problem using backtracking ? Solve the following instance

    $S = \{3, 7, 10, 5\}$, $d = 10$.

14. Write short notes on : **(4+3+3)**

    i) Merge sort.

    ii) Problem reduction.

    iii) P and NP problems.

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(CBCS) (2020 – 21 & Onwards)
COMPUTER SCIENCE
2MCA5 : The Design and Analysis of Algorithms

Time : 3 Hours Max. Marks : 70

Instructions *: 1) Section – **A** : Answer **any five** questions.*
*2) Section – **B** : Answer **any four** questions.*

SECTION – A

Answer **any five** questions. **Each** question carries **six** marks. **(5×6=30)**

1. Explain various steps involved in Algorithm Problem solving.
2. Explain different asymptotic notations in detail.
3. Discuss the Brute-Force string matching algorithm.
4. Present the steps used in Mathematical Analysis of non-recursive algorithm with example.
5. What is Hashing and Hash Function ? Explain various methods of choosing a Hash function.
6. Explain the Quick Sort Algorithm with its time complexity.
7. Compare and contrast Divide and Conquer Vs Dynamic Programming.
8. Explain the steps to solve sum of subsets problem using Backtracking.

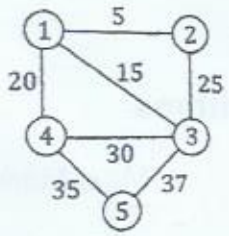
SECTION – B

Answer **any four** questions. **Each** question carries **10** marks. **(4×10=40)**

9. a) Explain Basic Efficiency Classes used in analysis of algorithm. **(4+6)**
   b) Apply Selection Sort technique to sort the given list of numbers : 45, 21, 52, 61, 9, 15.

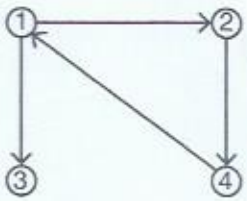
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10. a) Trace the Prim's Algorithm for the following graph. **(6+4)**



b) Give an account of Topological Sorting with the methods to achieve Topological sorting.

11. Given a Digraph, how do you arrive at its transitive closure ? Apply Warshall Algorithm to compute the transitive closure of the given graph. **10**



12. a) Explain the steps involved in Sorting by Counting. **(4+6)**

b) Apply Horspool's algorithm to search for pattern "BROWN" in the text "THATCOLOURISNOTBROWN".

13. Define N Queens problem. Explain 4-Queens problem and draw its solution space tree.

14. Write a short notes on : **(3+3+4)**

i) Breadth-First Search

ii) Traveling Salesman Problem

iii) Strassen's Matrix Multiplication.

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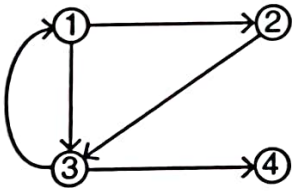
II Semester M.C.A. Examination, December 2024
(2020 – 21 and Onwards) (CBCS)
COMPUTER SCIENCE
2MCA5 : The Design and Analysis of Algorithms

Time : 3 Hours Max. Marks : 70

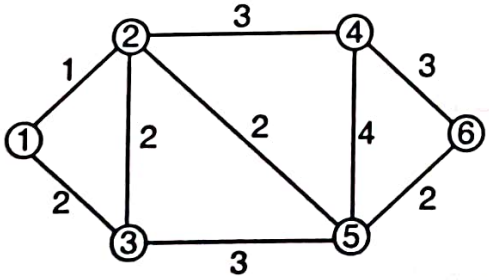
Instruction : *Answer* ***any five*** *from Part –* ***A*** *and* ***any four*** *from Part –* ***B****.*

PART – A **(5×6=30)**

1. Define Big oh. Write an algorithm to solve the below problem. Find its complexity.
   Sum = $1^2 - 3^2 + 5^2 - 7^2 + 9^2 + ... + n^2$.
2. Write insertion sort algorithm. Sort the following numbers using insertion sort algorithm :
   {9, 4, 6, 3, 7}
   What is the best case for insertion sort ?
3. Why is Strassen's matrix multiplication algorithm is better than brute force matrix multiplication algorithm ? Find complexity of Strassen's matrix multiplication algorithm.
4. What is hashing ? How is hash search better than brute force searching algorithm ?
5. What are P, NP and NP-complete problems ? Give examples.
6. Write an algorithm for input enhancement string matching. Find its complexity.
7. Find transitive closure of a graph using warshall algorithm for the following graph. (Show the matrix step by step)



8. Find the minimum shortest path using Prim's Algorithm.



P.T.O.

PART – B (4×10=40)

9. Write algorithm for Merge sort. Find its complexity. Trace your algorithm to sort the following numbers : {4, 9, 3, 6, 2, 5}

10. Given the following frequency chart, find Huffman's tree.

Character	i	a	♭	m	f	o	l
Frequency	0.15	0.2	0.2	0.15	0.1	0.05	0.15

Find the encoding for i♭am♭a♭fool.

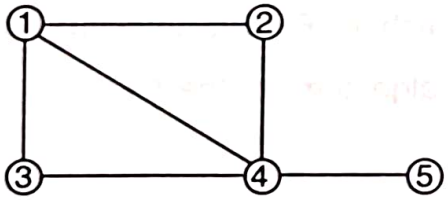
11. Find $^{7}C_{4}$ using dynamic programming. Find complexity to find $^{n}C_{p}$.

12. Solve the following assignment problem using Branch and Bound method.

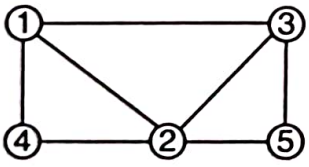
Person \ Job	1	2	3	4
A	2	4	2	5
B	1	3	3	1
C	4	6	5	3
D	2	5	4	6

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13. Find DFS and BFS for the following. What is the complexity of these algorithms ?



14. Find the Hamiltonian path for the following.



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(CBCS) (2020-21 and Onwards)
COMPUTER SCIENCE
2MCA5 : The Design and Analysis of Algorithms

Time : 3 Hours

Max. Marks : 70

Instructions : *1) Section – **A** : Answer **any five** questions.*
*2) Section – **B** : Answer **any four** questions.*

SECTION – A

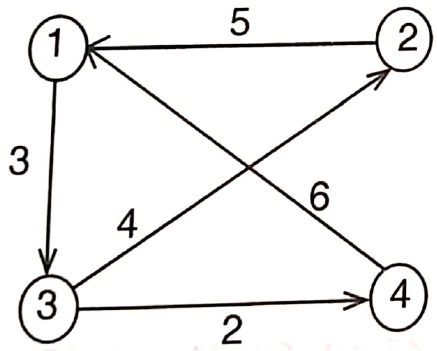
Answer **any five** questions. **Each** question carries **six** marks : **(5×6=30)**

1. Present Brute-force based algorithm to perform pattern matching and investigate its efficiency.

2. Explain different asymptotic notations in detail.

3. Explain the towers of hanoi algorithm and find its solution i.e. no. of moves required for n disks [Analyze the recursive algorithms].

4. Solve all pairs shortest path problem for the below graph using Floyd's algorithm.

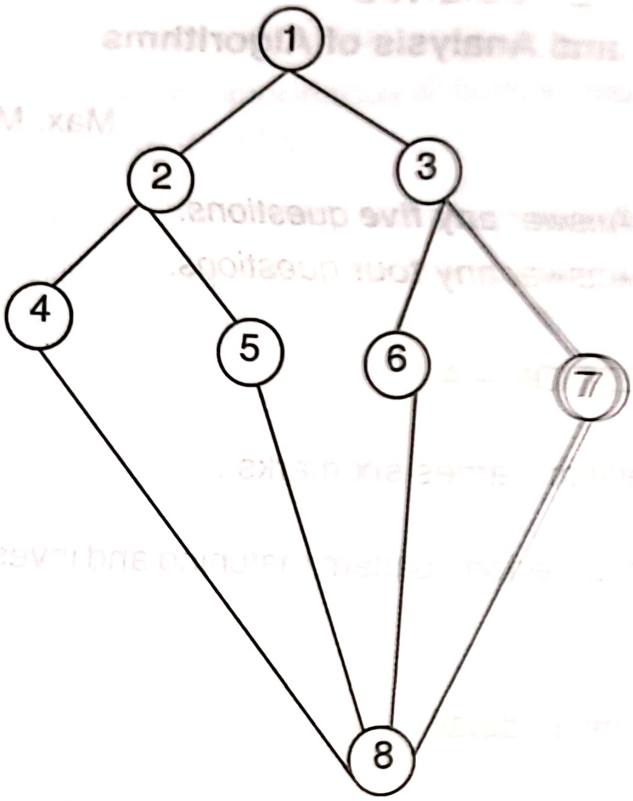


5. Write the quick sort algorithm and find its time complexity [use the following numbers] (Assume the first element as pivot element)
35, 36, 15, 45, 92, 71, 9, 33

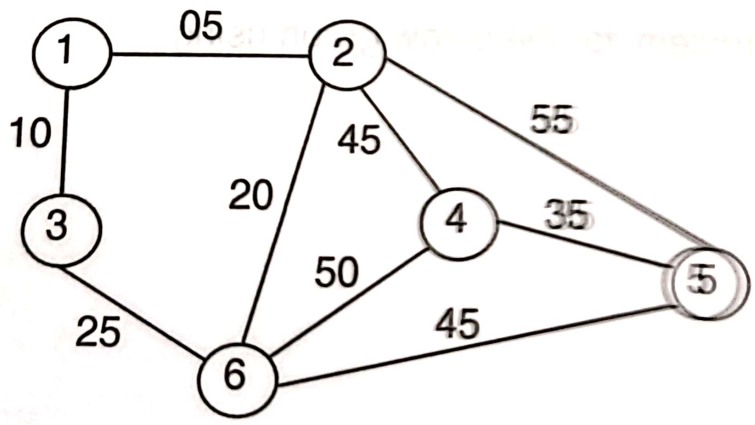
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6. Perform DFS and BFS traversals for the following. What is the complexity o these algorithms ? (start with node 1)



7. Find the minimum weight spanning tree by Kruskal's algorithm.



8. Apply selection sort technique for the given list 45, 21, 56, 71, – 1, 15 and find its time complexity.

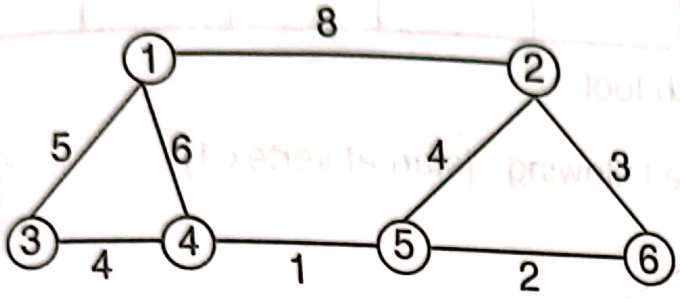
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SECTION – B

wer **any four** questions. **Each** question carries **ten** marks : (4×10=40)

) Trace the Prim's algorithm for the following graph. 6

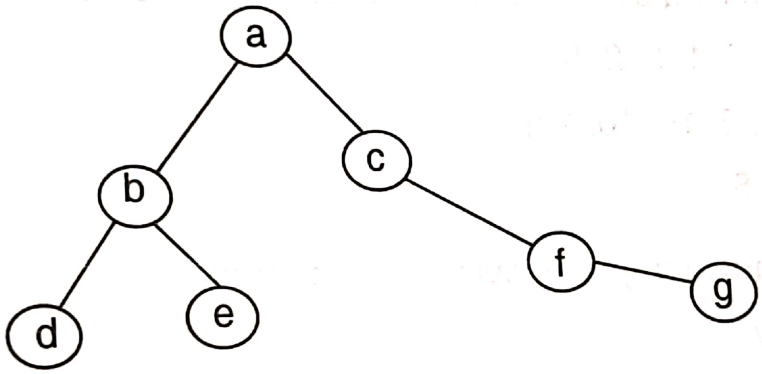


) Perform the following traversals on the given binary 4

a) preorder

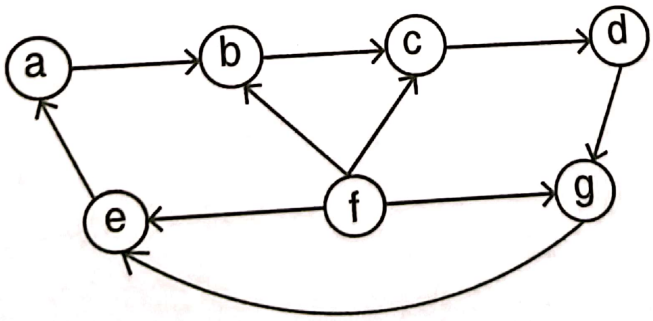
b) inorder

c) post order.



Which one yields a sorted order list of applied to binary search tree ?

a) Find the solution to topological sorting problem using DFS-based algorithm for the below graph. 5



b) Find 7_{C_4} using dynamic programming.

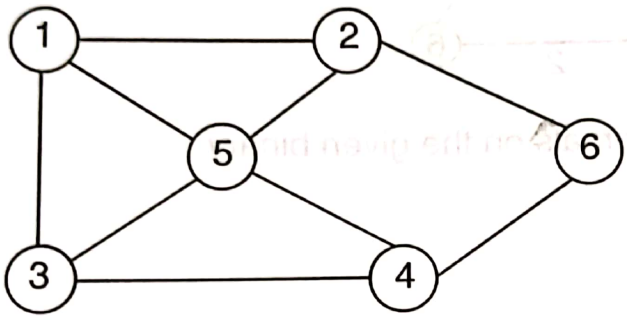
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11. Given the following frequency chart, find Huffman's tree. 10

Character	i	a	␣	m	f	o	l
Frequency	0.15	0.2	0.2	0.15	0.1	0.05	0.15

Find the encoding for i␣ am␣a␣ fool.

12. Find the Hamiltonian path for the following : [start at vertex 1] 10



13. Using dynamic programming solve the following Knapsack problem. 10

$[W_1, W_2, W_3, W_4, W_5] = [2, 1, 5, 2, 5]$

$[P_1, P_2, P_3, P_4, P_5] = [20, 5, 15, 10, 12]$

Knapsack capacity M = 8.

14. a) Advantages and disadvantages of divide and conquer. 4

b) Principle of optimality. 3

c) Advantage and disadvantage of depth-fast search. 3

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III Semester M.Sc. Examination, May 2024
(CBCS Scheme)
COMPUTER SCIENCE (2022 – 2023 Onwards)
MSC303 : Design and Analysis of Algorithms

Time : 3 Hours Max. Marks : 70

Instructions : *1) Part – **A** : Answer **any five** questions.*
*2) Part – **B** : Answer **any four** questions.*

PART – A

Answer **any five** questions. **Each** carries **six** marks : **(5×6=30)**

1. What are the fundamentals of Algorithmic problem solving ?
2. Discuss about asymptotic notations and basic efficiency classes.
3. Produce Brute-Force string matching algorithm.
4. Give an example to explain the Horspool's algorithm and construct a shift table for it.
5. Explain P and NP problems with example.
6. Generate all permutations of {1, 2, 3, 4} by backtracking.
7. Explain about parallel sorting networks.
8. What are the applications on graph problems ?

PART – B

Answer **any four** questions. **Each** carries **ten** marks : **(4×10=40)**

9. a) Explain about the general plan for analyzing time efficiency of nonrecursive and recursive algorithms. **6**

   b) Using the definition, design an algorithm that computes the nth Fibonacci number recursively. **4**

P.T.O.

10. a) Discuss about binary tree traversals and related properties. **6**

    b) Write a pseudocode for obtaining the product of two matrices using the Strassen's matrix multiplication. **4**

11. a) What is hashing ? Explain about open hashing and closed hashing. **6**

    b) Produce Huffman's algorithm. **4**

12. a) Write a short note on Knapsack problem. **4**

    b) Explain about traveling salesman problem with an example. **6**

13. a) Explain about Monte Carlo algorithm. **6**

    b) Discuss about Game-theoretic techniques. **4**

14. Write a short notes on :

    a) Algorithm visualization. **5**

    b) Topological sorting. **5**
